

Block Diagram — End-to-End Retrieval Architecture

Documents → Chunks → Embeddings → Index → Retrieval → LLM / Agent

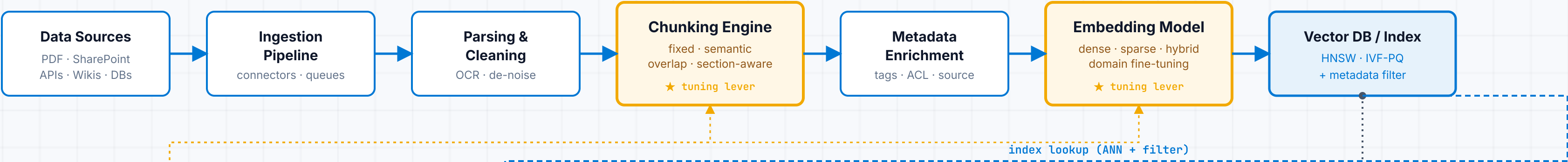
LEGEND

Ingestion path

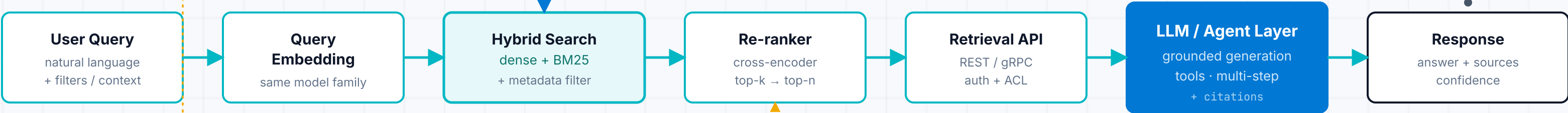
Query / retrieval path

Tuning & control plane

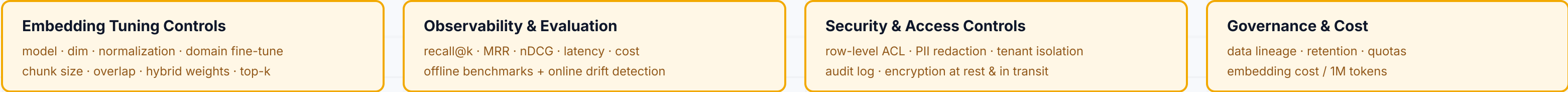
INGESTION ▶



QUERY ▶



CONTROL PLANE ▶



SYSTEM FLOW

Five stages from raw corpora to grounded answers — chunking and embeddings are the quality levers.

① **Ingest & Parse**
Connectors normalize PDFs, SharePoint, APIs into clean text + structure.

② **Chunk & Embed**
Chunking strategy & embedding model jointly drive retrieval precision.

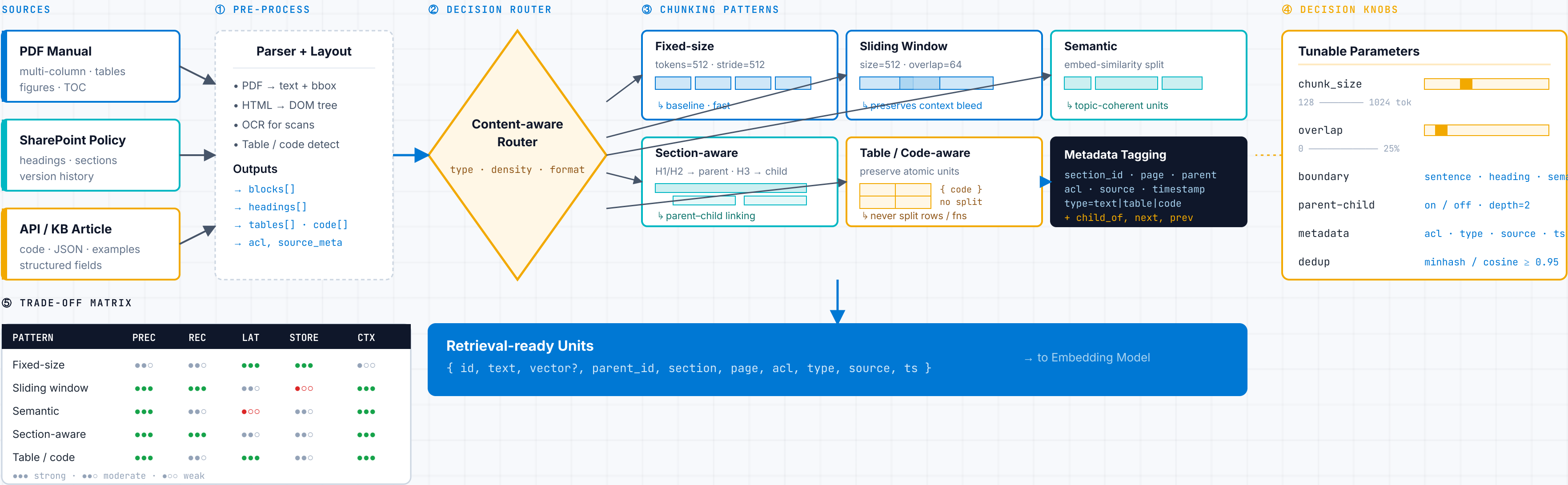
③ **Index**
ANN index (HNSW / IVF-PQ) with metadata filters and tenant scoping.

④ **Hybrid Search + Re-rank**
Dense + BM25, cross-encoder re-rank for top-N relevance.

⑤ **Ground LLM**
Agent grounds answers in retrieved chunks with full citation trail.

Chunking Strategy Schematic

Raw documents → retrieval-ready units · patterns, decision points, trade-offs



Embeddings Tuning — Wiring Diagram

Ten-stage runtime design with separate data, tuning, and evaluation feedback loops

LOOP LEGEND

Data flow (forward)

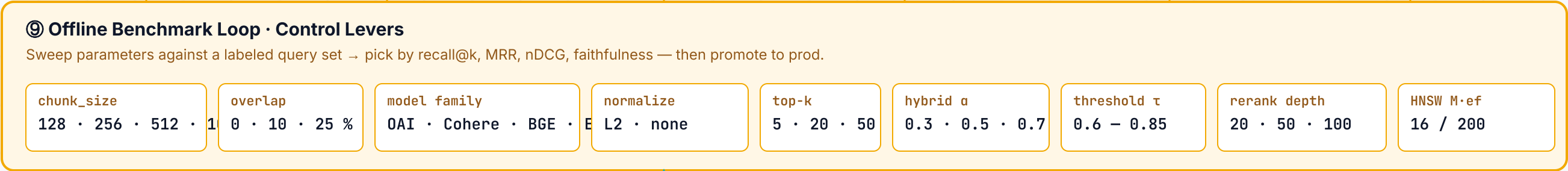
Tuning loop (offline)

Evaluation feedback

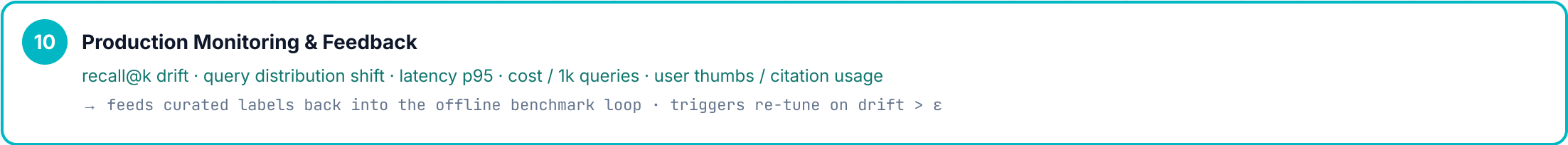
DATA FLOW ▶



TUNING LOOP ▶



EVAL FEEDBACK ▶



OPTIMIZATION GOALS

◆ Retrieval accuracy (recall@k, nDCG)

◆ Latency p95 (sub-300 ms end-to-end)

◆ Cost efficiency (\$ / 1k queries)

◆ Explainability (citations, audit)

ENTERPRISE CONCERNS

- 🌐 Multilingual Corpora

Pick multilingual embeddings or train language adapters; never mix incompatible model families across the index.
- 📖 Domain Terminology

Augment with domain glossaries, fine-tune on labeled in-domain pairs, or layer a sparse BM25 channel for exact terms.
- 📉 Embedding Drift

Monitor query / corpus distribution; schedule scheduled re-embed jobs when model version or corpus topics shift.
- 🛡️ Governance & Audit

Version every model + index; retain query / retrieval logs with citations for regulatory and red-team review.

PRODUCTION KPI TARGETS

RECALL @ 10

≥ 0.92

on labeled eval set

FAITHFULNESS

≥ 0.95

LATENCY P95

< 300 ms

retrieval + rerank

COST / 1K QUERIES

< \$0.40